



**Food and Agriculture
Organization of the
United Nations**

Hidden costs of potato and rice production

**Insights from a survey-based true cost accounting analysis
in Bhutan, Burkina Faso and Malawi**



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Abbreviations

DALY	disability-adjusted life years
LCA	life cycle assessment
LCIA	life cycle impact assessment
PPP	purchasing power parity
TCA	true cost accounting
SRI	System of Rice Intensification

1 Introduction

Agrifood systems provide large benefits in terms of food security and employment worldwide. At the same time, they are responsible for huge socioeconomic and environmental damages. Recent estimates indicate that global quantified hidden costs of agrifood systems exceeds USD 10 trillion a year (about 10 percent of global GDP PPP in 2020) (FAO, 2023a; Ruggeri Laderchi *et al.*, 2024).¹ Uncovering and quantifying hidden cost of agrifood systems – both at the aggregate and product level – is a first key step to generate information needed by stakeholders to transform agrifood systems and making them more sustainable. True cost accounting (TCA), “a holistic and systemic approach to measure and value the environmental, social, health and economic costs and benefits generated by agrifood systems” (FAO, 2023b, p. xv) offers an important pathway towards achieving this goal by revealing and clarifying hidden costs to policymakers, businesses, farmers, investors, and consumers. However, data scarcity and lack of capacity are two major barriers to scaling up TCA adoption, especially in middle- and low-income countries (FAO, 2023b, p. 76).

The study aimed at contributing to *The State of Food and Agriculture 2024* by showcasing how TCA can be used in middle- and low-income countries to identify hidden costs by conducting a TCA targeted assessment of different rice and Irish potato value chains in Bhutan, Burkina Faso and Malawi. The main objectives of the research were: i) to identify what are the environmental and socioeconomic hidden costs hot spots of the targeted value chains and ii) to provide insights and lessons learned concerning survey-based TCA in low and middle-income countries, which can be useful for future TCA targeted assessments in similar contexts. Although to a different extent, rice and potato play a key role as source of revenue, income diversification, and in diets in the selected countries (Cromme *et al.*, 2010; D’Alessandro and Tondel, 2021; FAO, European Union and CIRAD, 2022, 2023; Kapalasa *et al.*, 2022). Understanding the hidden costs associated to the two crops can therefore contribute to informing policies to strengthen rice and potato value chains and make them more sustainable from both environmental and social perspectives.

¹ A hidden cost is “any cost to individuals or society that is not reflected in the market price of a product or service. It refers to external costs (that is, a negative externality) or economic losses triggered by other market, institutional or policy failures.” (FAO, 2023a, p. xi).

2 Methodology: overview

The study was conducted in four main phases. The first phase involved mapping of the value chains. In the second phase, indicators to include in the TCA assessment were selected and questionnaires for primary data collection were developed. The third phase focused on data collection, both primary and secondary. In the final phase the selected indicators of hidden costs were calculated.

2.1 Mapping of potato and rice value chains

Mapping of the value chains in the countries was conducted with support of the local research centres. It involved identifying the value chain actors, agricultural practices (including irrigation), crop varieties, and the cultivation areas. To distinguish the specific value chains the following criteria were considered: the type of market channels (smallholders vs commercially oriented farmers), input use (conventional vs organic), rice management practices (conventional vs System of Rice Intensification [SRI]), or water regime (rainfed vs irrigation). The mapping resulted in isolating ten and nine distinct value chains suitable for the TCA assessment for potato and rice, respectively. For potato, the following value chains were isolated: in Bhutan “smallholders”, “commercial” and “organic”; in Burkina Faso “conventional small and medium producers”, “conventional large”, “conventional high-input” and “organic”; in Malawi “low-input subsistence, rainfed”, “local varieties, mineral fertilizer, irrigated” and “improved varieties, mineral fertilizer, irrigated”. For rice, the following value chains were isolated: in Bhutan “irrigated, subtropical and temperate”, “rainfed” and “organic”; in Burkina Faso “conventional rainfed”, “conventional irrigated”, “SRI rainfed” and “SRI irrigated”; in Malawi “low-input subsistence, rainfed” and “conventional irrigated”.²

2.2 Indicators and questionnaire development

The selection of indicators followed the general TCA framework, in which agrifood systems activities depend on, and at the same time affect, natural, human, social and produced capitals (FAO, 2023a; TEEB, 2018). This study focused on measuring hidden costs related to natural and human capitals for the identified production systems. The specific indicators and monetisation methods were based on recent literature (Eingenraam *et al.*, 2020; True Cost Initiative, 2022).³

To estimate natural capital related hidden costs a cradle-to-farm gate life cycle assessment (LCA) was conducted following the structure of the ISO standards 14040 and 14044 and using the 15 impact categories of the life cycle impact assessment (LCIA) method Recipe 1.13 Midpoint (Hierarchist). The selected functional unit (FU) is 1 kg of rice and potato, respectively.⁴

The monetization factors for the environmental impacts are based on the Product Environmental Footprint (PEF) methodology of the European Commission (Zampori and Pant, 2019) and the monetization approach described in the report prepared by Trinomics (European Commission, Directorate General for Energy *et al.*, 2020), which specifies the external costs for the respective impact categories. The impact categories assessed as well as the respective monetization factors are shown in Table 1.

Further information on the LCIA methodology used in the study are provided in Annex 2.

² The value chains are presented more in detail in section 3 along with the associated descriptive statistics.

³ Social and produced capitals were outside the scope of the impact assessment of the study. Social capital refers to “Networks, including institutions, together with shared norms, values and understandings that facilitate cooperation within or among groups” (FAO, 2023a, p. xi). Produced capital refers to “All manufactured capital, such as buildings, factories, machinery and physical infrastructure (roads, water systems), as well as all financial capital and intellectual capital (technology, software, patents, brands and so on)” (FAO, 2023a, p. xi).

⁴ Depending on the objective of the analysis, a land-based functional unit (e.g. 1 hectare) may also be considered when assessing environmental impacts as it would allow to account for absolute level of emissions/impacts. The importance of the choice of functional unit and its implications are discussed more in detail in Section 4.4.

Table 1. Environmental impact categories and monetization factors

Environmental impact category	Unit	Monetization factor (2023 PPP dollars)
Acidification	Mol H ⁺ -eq	5.74E-01
Climate change	Kg CO ₂ -eq	1.71E-01
Ecotoxicity, freshwater	CTUe	6.37E-05
Eutrophication, freshwater	Kg P-eq	3.20E+00
Eutrophication, marine	Kg N-eq	5.36E+00
Human toxicity, cancer	CTUh	1.51E+06
Human toxicity, non-cancer	CTUh	2.73E+05
Ionising radiation	kBq U235-eq	2.00E-03
Land use	Dimensionless	2.92E-04
Ozone depletion	Kg CFC-11-eq	5.24E+01
Particulate matter	Disease incidence	1.31E+06
Photochemical ozone formation	Kg NMVOC-eq	1.99E+00
Resource use, fossils	MJ, Net calorific value	2.17E-03
Resource use, minerals and metals	Kg Sb-eq	2.74E+00
Water use	m ³ world eq. deprived	8.33E-03

Source: Authors' own elaboration based on European Commission & Trinomics. 2020. *External costs – Energy costs, taxes and the impact of government interventions on investments – Final report*. Brussels, European Commission. <https://data.europa.eu/doi/10.2833/81390>

Human capital hidden costs indicators were selected considering key dimensions of decent rural employment such as income, working conditions, and workers' health (FAO, 2017) using as reference the *True cost accounting agrifood handbook's* guidelines developed by the True Cost Initiative and monetisation factors from Sustainability Impact Metrics (Sustainability Impact Metrics, 2024; True Cost Initiative, 2022)⁵. Table 2 provides an overview of the socioeconomic impacts assessed in the analysis.

For the assessment of the income levels and its impact on local livelihoods the i) living income gap, referring to the household income required for a decent life in the respective country and ii) potential differences in the remunerations of female and male workers (Gender Pay Gap) were considered (Anker and Anker, 2017; Cruickshank and Bammann, 2024; Sustainability Impact Metrics, 2024; True Cost Initiative, 2022).

The working conditions are examined based on the indicators aimed at capturing the potential occurrence of Excessive Working Hours, Unpaid Labour and Child Labour. The benchmark set for excessive working is more than 2 240 hours per year (Sustainability Impact Metrics, 2024; True Cost Initiative, 2022). Unpaid labour is proxied by on-farm work of adult non-household members performed without remuneration.

As child labour has been documented in the agricultural sectors of Bhutan, Burkina Faso (in particular in the cotton sector), and Malawi (tea and tobacco) (FAO, 2023b; Government of Malawi, 2023; ILO, 2021, 2023) we proxied for employment of children in the potato and rice value chains. Typically, in agriculture work peaks occur for instance at times of sowing and at harvest (Sustainability Impact Metrics, 2024). Therefore, children helping their parents occasionally during work peaks, whereby education and mental and physical development are secured, is considered not that severe as child labour during the transport, processing, and marketing of agricultural commodities (Sustainability Impact Metrics, 2024). Hence, the potential occurrence of child labour is proxied by working hours in farming above 560 hours per year of children below 15 years old (Sustainability Impact Metrics, 2024; True Cost Initiative, 2022).

⁵ Decent rural jobs, defined as "...opportunities for work that are productive, respect core labour standards, provide a fair income (whether through self-employment or wage labour) and ensure equal treatment for all. Workers should be able to perform their tasks under safe and healthy conditions and have a voice in the workplace" (FAO, 2017, p. 4).

Agricultural workers fulfil various tasks under outdoor conditions. It is one of the sectors with most occupational health and safety threats including heavy loads, loud noises, exposure to harmful substances like pesticides and rough terrains (FAO, 2017; True Cost Initiative, 2022). Therefore, the occurrence of injuries, illnesses and fatalities related to the work performed by farmers is analysed and the costs caused by work-related injuries, illnesses and fatalities are taken into account through the indicator Occupational Safety and Health.

In order to make the results of the three countries comparable, the purchasing power parity (PPP) were calculated and all costs and prices were stated in 2023 PPP international dollars using PPP conversion factors from International Monetary Fund (IMF)'s *World Economic Outlook* (IMF, 2024).

Additional information and formulas of the indicators are provided in Annex 3.

Table 2. Socioeconomic indicators

Impact category	Description	Unit	Monetization factor (2023 PPP dollars)
Living income gap	Difference between the living income in the respective country and the actual income.	2023 PPP international dollar per working hour	
Gender pay gap	Difference between male and female contribution to the household income.	2023 PPP international dollars per working hour	
Excessive working hours	Annual average working hours above 2 240 hours.	Disability-adjusted life years (DALY)	145 531/DALY
Unpaid labour	Annual average on-farm working hours of non-household member performed without remuneration per national minimum wage.	2023 PPP international dollars per working hour	
Child labour	Annual average working hours above 560 hours of children below 15 years old.	DALY	145 531/DALY
Occupational safety and health	Work related injuries, (long term) illnesses and fatal cases.	DALY	145 531/DALY

Source: Authors' own elaboration based on True Cost Initiative. 2022. *True cost accounting agrifood handbook – Practical guidelines for the food and farming sector on impact measurement, valuation and reporting*. https://tca2f.org/wp-content/uploads/2022/03/TCA_Agrifood_Handbook.pdf

The selection of indicators allowed for defining primary and secondary data needs. Once the primary data needs were identified, a standardized questionnaire for each crop was developed. To adequately collect data for the many dimensions needed for the TCA assessment, the questionnaire included the following sections: cultivation, storage, packaging, processing, transport, retail and socioeconomic aspects. The questionnaires were subsequently adapted to each country context: beyond the inclusion of local measurement units, answer options were also modified to allow for the selection of local type of fertilizer and other inputs, tools, and crop varieties. Further, different types of actors were considered: farmers, non-household farm workers, processors, transporters, retailers and key informants (e.g. non-governmental organizations [NGOs], government officials).

2.3 Data collection

Given the purpose of conducting a pilot TCA assessment, this study used a non-random convenience sampling approach. As a result, our sample should not be considered representative, and the results should not be used for drawing any policy recommendations.

The data collection efforts primarily concentrated on the production stage of value chains. This focus was chosen because it was feasible to gather a sufficient number of observations exclusively at this stage, involving actors such as farmers, their household members, and non-household farm workers.

Additional interviews were also conducted with actors involved in other stages of the value chain, including processors, transporters, retailers, government officials and NGOs to validate the findings from primary and secondary data collection.

Primary data were collected between July and October 2023 by the local research centres using tablets and the Survey Solutions software. In Bhutan, 124 and 138 farmers were interviewed for potato and rice, respectively, across the following districts: Chukha, Gasa, Paro, Punakha, Sarpang, and Wangdue. In Burkina Faso, 123 potato farmers and 123 rice farmers were interviewed. Survey sites for potato included: Ouahigouya, Tougzagué, Niéssega, and Karangasso Sambla. Rice farmers were interviewed in: Komki-Ipala, Kouzoughin, Goghin, Boulbi, Samandeni, Tanwogoma, Bama. In Malawi, 117 potato farmers were interviewed in the Dedza and Ntcheu districts and 90 rice farmers in Karonga and Nkhata Bay districts.

Some challenges arose during the data collection process, which affected the final sample size. In Burkina Faso, some large scale potato producers that were initially planned to be interviewed fled their areas due to security issues. As a result, it was possible to interview only 12 of them. In Malawi, only seven farmers using improved potato varieties were identified, a result in line with the difficulties related to the potato seed system in Malawi, characterized by the predominant use by farmers of poor-quality seeds (Kapalasa *et al.*, 2022). In addition, many farmers could not provide estimates for some of the variables, such as water and energy usage as they often do not keep records of (some) inputs used in the production process or unable to probe or recall them.

After data collection was completed, workshops were held in each country from October to November 2023. These sessions were designed to present and validate the results of the data collection for the identified value chains. Additionally, the workshops provided an opportunity to gather further insights and information from local stakeholders.

Following the completion of the surveys in the three countries, and after the validation workshops took place, and the collected data was prepared for both the LCA and socioeconomic indicators assessment. The analysis relied almost entirely on the collected primary data, however secondary data were also used for the LCA and for the calculation of some socioeconomic indicators. More detailed on the calculations are provided in Annexes 2 and 3.

2.4 Evaluation of hidden costs

Once completed the data cleaning process for both the socioeconomic data and the life cycle inventories for the LCA, all the indicators introduced in Sections 2.2 were calculated for each selected value chain. The obtained values were then multiplied by the respective monetization factors to obtain the estimates of the hidden costs, and subsequently compared and analysed.

3 Descriptive statistics

This section provides an overview of the value chains and associated key descriptive statistics. As the data collection was mainly targeting production stage of the value chains, the descriptive statistics report the farm-level data.

In Bhutan, smallholder production is characterized by an average cultivation area of 0.45 ha and yield of 8.6 tonnes/ha (Table 3). Commercially oriented farmers cultivate slightly larger areas and have higher yields with respect to smallholders, on average 0.68 ha and 15.9 tonnes/ha, respectively. Organic farmers include smallholders who do not use any mineral fertilizer and have average yield of 4.9 tonnes/ha.

Potato systems in Burkina Faso were grouped in four main systems based on average plot size and use of inputs: small and medium, large, high-input conventional producers and organic producers. Organic producers, beyond not using mineral fertilizers, are characterized by small plots and high rate of nitrogen fertilizers in comparison with the other production systems (0.07 ha and 529 kg N/ha) and they have relatively high productivity (17.2 tonnes/ha).⁶

In Malawi, farmers using improved potato varieties have on average higher yield (7.8 tonnes/ha) with respect to subsistence-oriented farmers and farmers using local varieties (2.9 and 5.1 tonnes/ha, respectively). Rainfed farmers have smaller plot sizes (average of 0.21 ha) with respect to potato farmers using irrigation (0.36 and 0.34 ha for conventional and improved varieties, respectively).

Table 3. Descriptive statistics of potato value chains

Country	Production system	Number of farmers	Average cultivation area (ha)	Average yield (tonnes/ha)	Irrigation	Average nitrogen fertilizer (kg of N/ha)
Bhutan	VC1: smallholders	25	0.45	8.6	Yes	205
	VC2: commercial	68	0.68	15.9	Yes	155
	VC3: organic	31	0.48	4.9	Yes	160
Burkina Faso	VC1: conventional small and medium	46	0.63	8.1	Yes	226
	VC2: conventional large	12	2.79	6.6	Yes	167
	VC3: conventional high input	35	0.63	9.4	Yes	299
	VC4: organic	30	0.07	17.2	Yes	529
Malawi	VC1: low input subsistence, rainfed	37	0.21	2.9	No	61
	VC2: local varieties, mineral fertilizer, extensive irrigation	74	0.36	5.1	Yes	99
	VC3: improved varieties, mineral fertilizer, irrigated	7	0.34	7.8	Yes	118

Note: VC = value chain.

Source: Authors' own elaboration.

In Bhutan, rice value chains are similar in terms of area and yield (Table 4). In general, rainfed systems have lower level of nitrogen fertilization (39 kg/ha) with respect to irrigated and organic producers (67 and 53 kg/ha).

⁶ Organic potato farmers sampled in this study were part of a cooperative of farmers in the Niessèga village, who share a total area of about 1.5 ha.

In Burkina Faso, the highest yield was found among irrigated SRI farmers, with 5 tonnes/ha on average. Average rice cultivation areas were slightly higher among rainfed farmers (0.78 and 0.86 ha for conventional and SRI, respectively) with respect to irrigated ones (0.56 and 0.89 ha).

In Malawi, two rice value chains were considered: low-input subsistence farmers dependent on rain for irrigation and conventional farmers using irrigation. The average cultivation area was similar across the two systems, while the largest yield was recorded among farmers using irrigation with an average of 3.2 tonnes/ha.

Table 4. Descriptive statistics of rice value chains

Country	Production system	Number of farmers	Average cultivation area (ha)	Average yield (tonnes/ha)	Irrigation	Average nitrogen fertilizer (kg of N/ha)
Bhutan	VC1: irrigated, subtropical and temperate	90	0.71	3	Yes	67
	VC2: rainfed	24	0.62	2.6	No	39
	VC3: organic	24	0.57	2.5	Yes	53
Burkina Faso	VC1: conventional rainfed	54	0.78	3.6	No	104
	VC2: conventional irrigated	31	0.56	3.9	Yes	142
	VC3: SRI rainfed	16	0.86	3.4	No	105
	VC4: SRI irrigated	22	0.69	5	Yes	93
Malawi	VC1: low-input subsistence, rainfed	65	0.55	2.5	No	58
	VC2: conventional irrigated	25	0.46	3.2	Yes	58

Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

4 Results

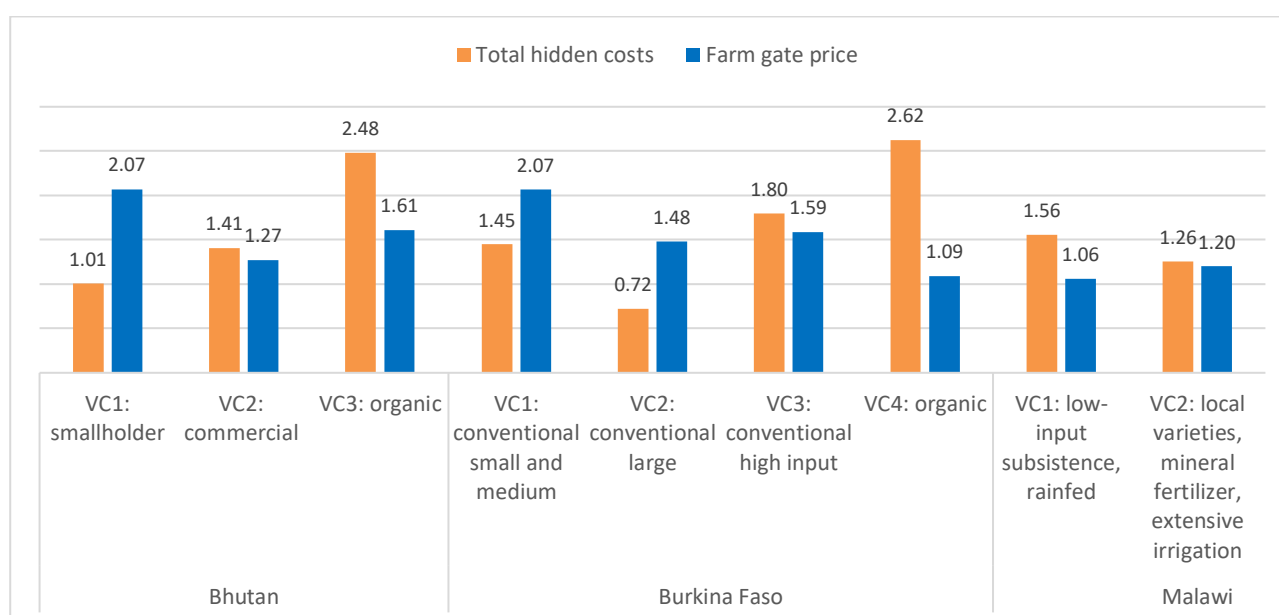
This section presents the results, starting from the main cost categories of the hidden costs (environmental and socioeconomic) and then discussing the impact indicators within each cost category. The indicators were calculated only at production level, given the focus of the data collection process.

4.1 Total hidden costs

The overall results show that the hidden costs for both potato and rice value chains are, on average, larger than the prices received by producers (Figures 1 and 2).

In potato production, the average estimates of total hidden costs value are about 1.2 times the farm gate price. However, substantial variations are found within countries. Total potato hidden costs range between 1.01–2.48, 0.72–36.2, and 1.26–2.62 dollars per kg, for Bhutan, Burkina Faso, and Malawi respectively.⁷ In particular, for Bhutan and Burkina Faso the highest ratio of total hidden cost to farm gate price is found in organic potato systems (VC3 Bhutan and VC4 Burkina Faso), with values of 1.5 and 24.6 respectively. The latter is due to very high socioeconomic hidden costs.

Figure 1. Total hidden costs and farm gate prices of potato value chains



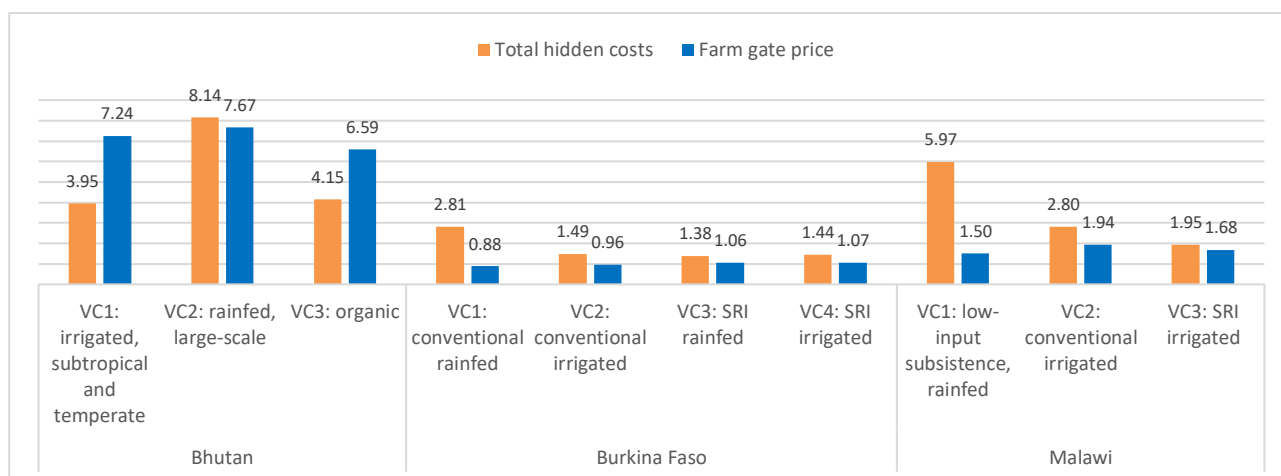
Note: VC = value chain.

Source: Authors' own elaboration.

In rice production, the average estimates of total hidden costs value are about 1.6 times the farm gate price. Similarly to potato systems, within country variations are found also in rice systems. Total rice hidden costs ranges are 3.95–8.14, 1.38–2.81 and 1.95–5.97 dollars per kg, for Bhutan, Burkina Faso, and Malawi respectively. In particular, in Bhutan the irrigated and organic systems have total hidden costs lower than farm gate prices, with ratios of hidden costs to farm gate prices of 0.54 and 0.62, respectively. Differently, in Malawi total hidden costs for rainfed low-input subsistence farmers are 3.98 times the farm gate price. In Burkina Faso, no large variation are observed among different systems with the exception conventional rainfed farmers, with a ratio of total hidden costs to farm gate price of 3.2, about three times higher than the other systems in the country. Overall, rainfed systems have the highest hidden costs within each country.

⁷ To enhance readability of Figure 1, VC4-potato Burkina Faso values were omitted from the figures. Organic potato total hidden costs and farm gate price are 36.23 and 1.47 (2023 PPP dollars), respectively. See Table A1 in the annex for absolute figures of organic potato hidden costs in Burkina Faso.

Figure 2. Total hidden costs and farm gate prices of rice value chains

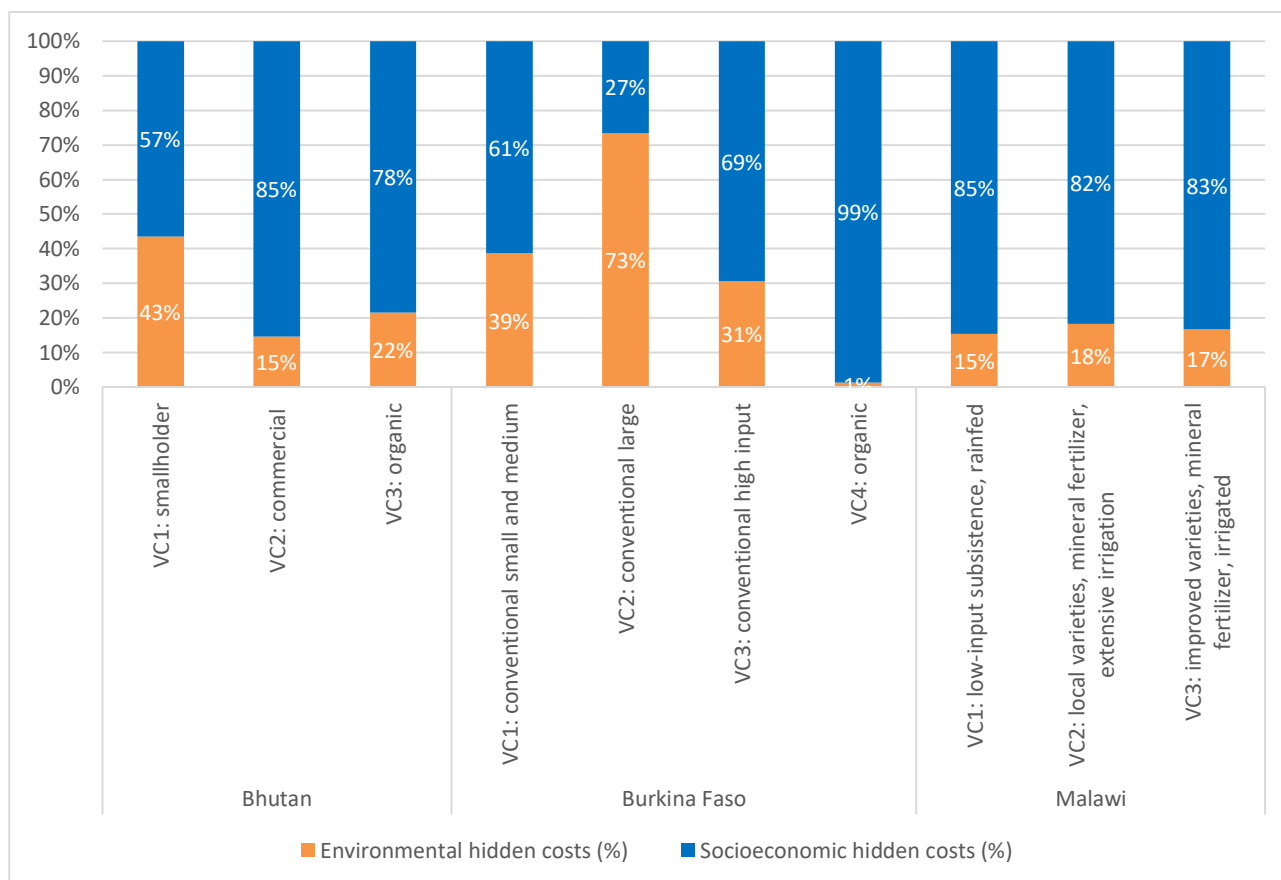


Note: VC = value chain.

Source: Authors' own elaboration

The majority of hidden costs is represented by socioeconomic costs for almost all potato (Figure 3) and rice (Figure 4) value chains. Socioeconomic cost shares range from a minimum of 27 percent of hidden costs (value chain [VC]2 potato Burkina Faso – conventional large producers) to a maximum of 99 percent (VC4 potato Burkina Faso – organic). Environmental hidden costs are the largest hidden cost category only for conventional potato production in Burkina Faso, representing 73 percent of hidden costs.

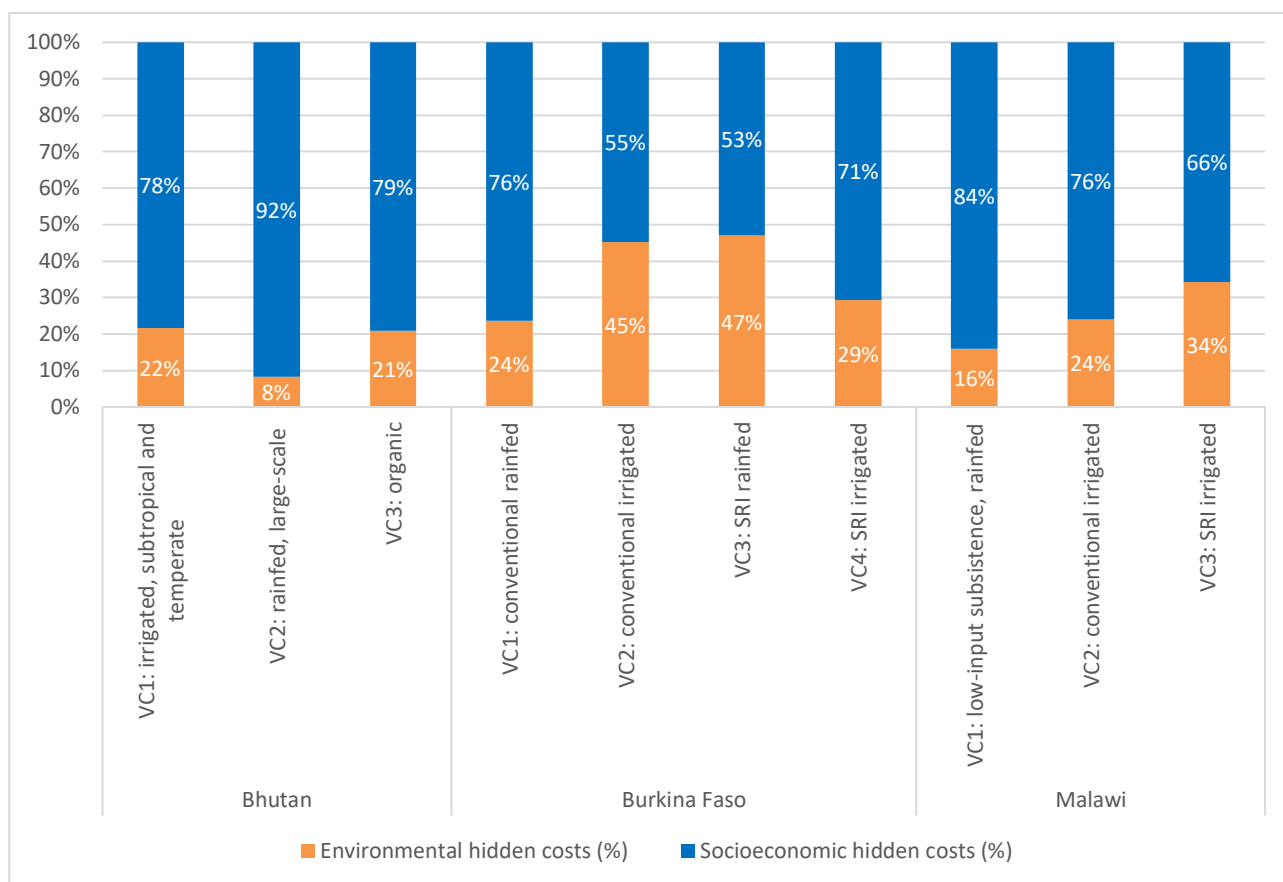
Figure 3. Share of hidden costs of potato value chains



Note: VC = value chain.

Source: Authors' own elaboration.

Figure 4. Share of hidden costs of rice value chains



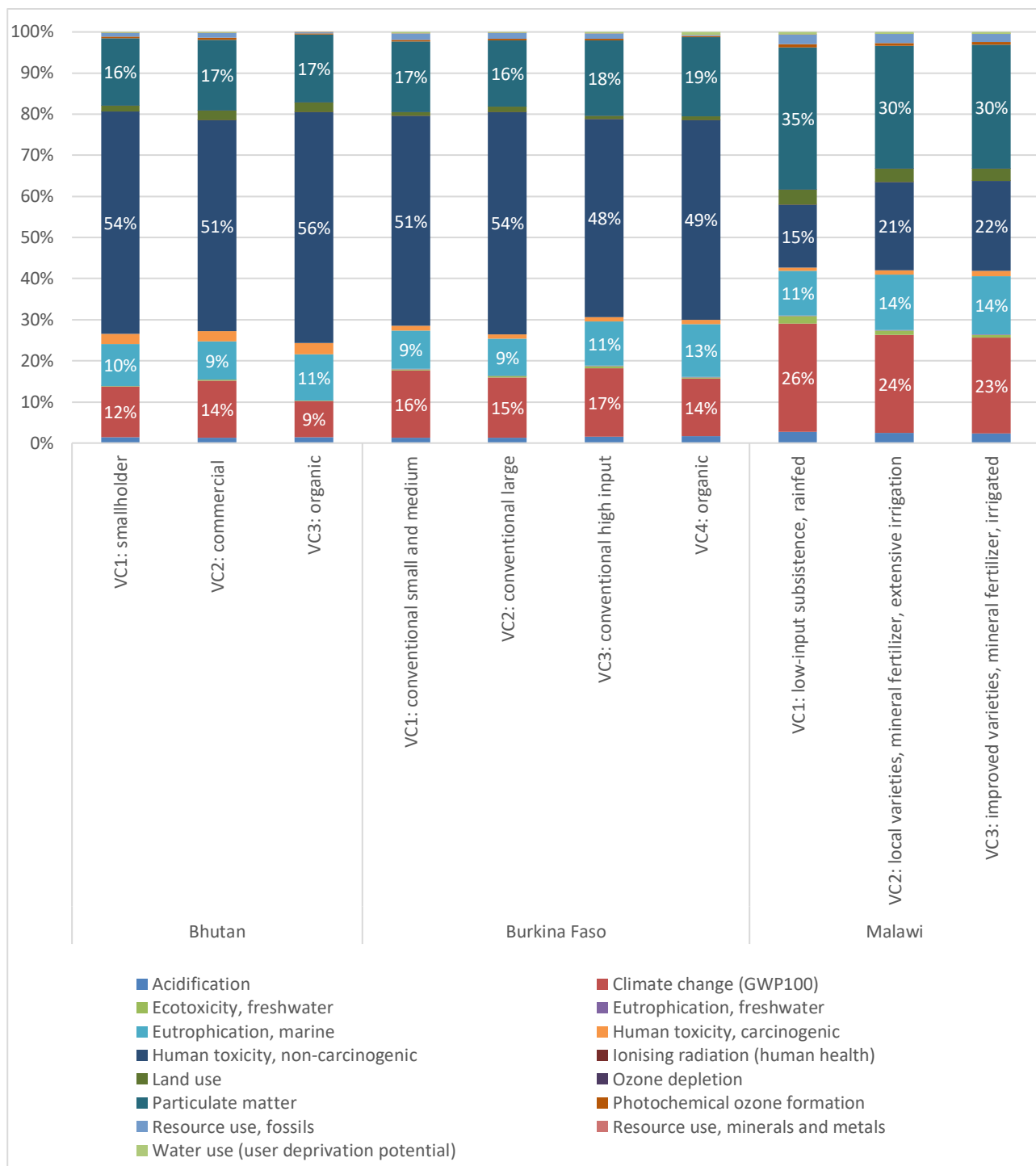
Note: VC = value chain.

Source: Authors' own elaboration.

4.2 Environmental hidden costs

In potato production, the environmental costs have different sources according to the producing country, while showing little variation within the same country (Figure 5). Human toxicity (non-carcinogenic) represents the main source of environmental costs in Bhutan and Burkina Faso (ranging from 48–56 percent of environmental costs), followed by particulate matter (16–19 percent) and climate change (9–17 percent). Differently, particulate matter is the main source of environmental costs in Malawi (30–34 percent), followed by climate change (23–26 percent) and human toxicity-non carcinogenic (15–22 percent).

Figure 5. Potato environmental hidden costs: shares of impact categories

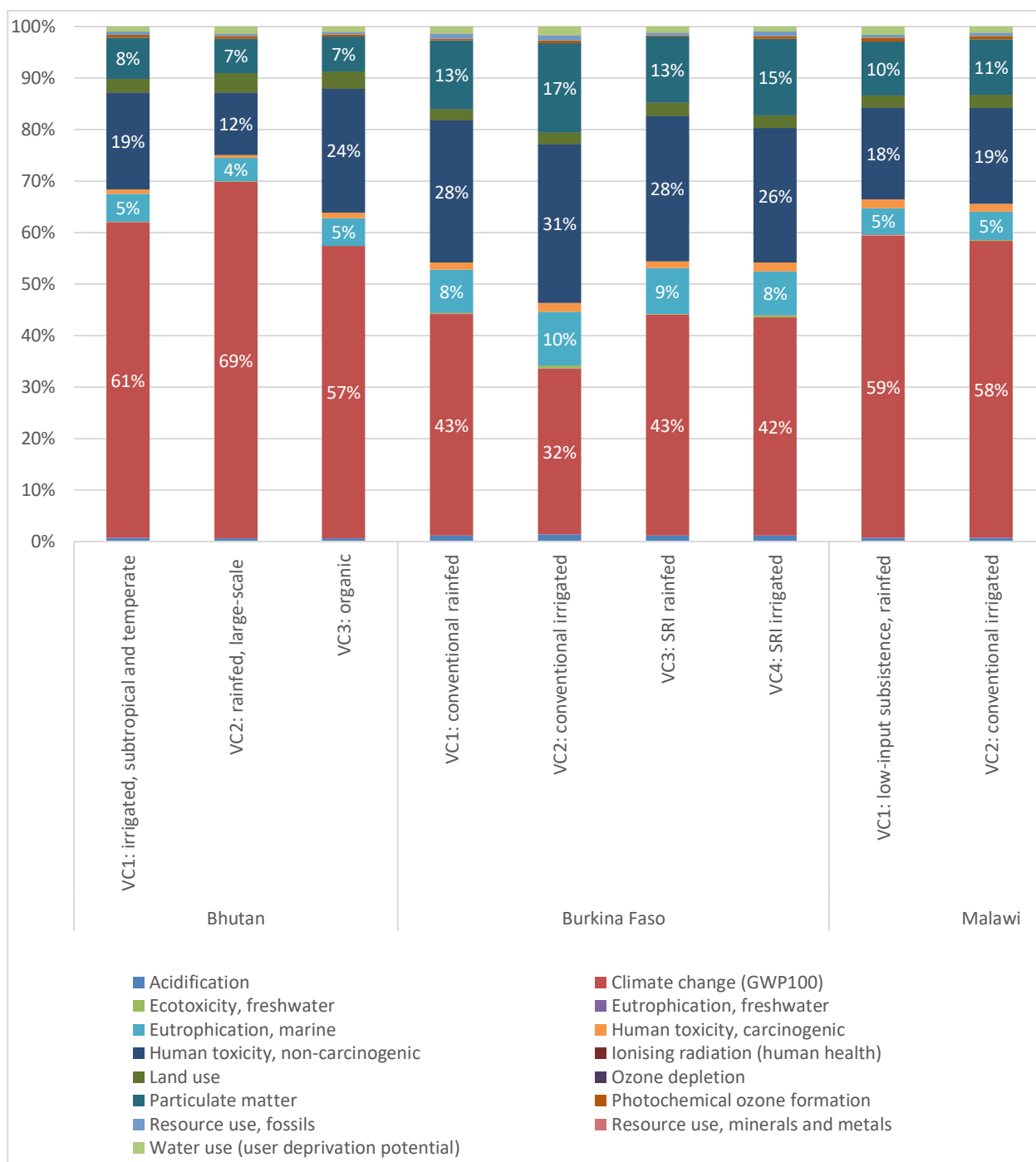


Note: VC = value chain.

Source: Authors' own elaboration.

For rice production, climate change is the main impact category, representing on average 60 percent of environmental hidden costs in Bhutan and Malawi, and about 40 percent in Burkina Faso. The second one is human toxicity (non-carcinogenic), particularly in Burkina Faso where its share is on average 28 percent. Overall, the relative shares of impacts in rice production do not vary substantially between different practices within the same country.

Figure 6. Rice environmental hidden costs: shares of impact categories



Note: VC = value chain.

Source: Authors' own elaboration.

For both potato and rice production, and in all three countries, human toxicity (non-carcinogenic), particulate matter and climate change are the three major contributors to the overall environmental hidden costs.

The impacts in the category human toxicity (non-carcinogenic) are mainly related to heavy metal emissions on the field. Heavy metals enter the system within the seeds, fertilisers and pesticides and accumulate within the product and the soil.

The main reason for particulate matter formation is ammonia emissions associated to nitrogen fertiliser application on the field and to open waste burning. The latter practice is particularly relevant in the context of Malawi, where waste disposal through open burning is a recognized issue (Government of Malawi, 2019).

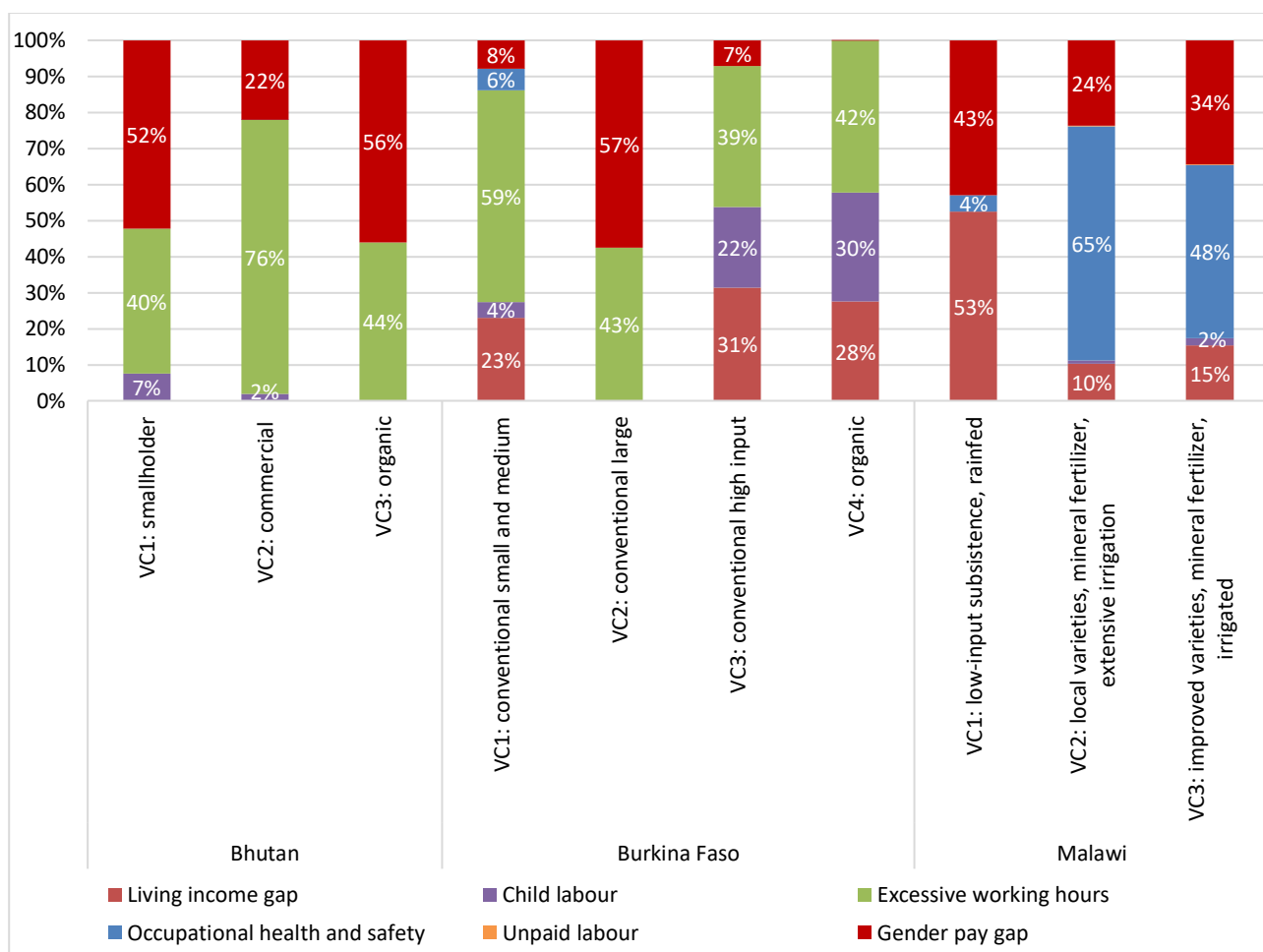
Climate change impacts in potato production are related to the energy-intensive production of fertilisers and the dinitrogen monoxide emissions on the field associated to fertiliser application. In addition, methane emissions contribute a large share of climate change impacts in rice production. Methane is emitted due to the anaerobic conditions in the rice fields and therefore depends on the flooding regime of the fields.

4.3 Socioeconomic hidden costs

Overall, the most important socioeconomic costs are excessive working hours, gender pay gap and living income gap. However, contrary to the environmental costs, socioeconomic costs are more differentiated both between and within countries as well as across crops.

Socioeconomic costs in potato production in Bhutan are mainly caused by excessive working hours – particularly in VC2 “commercial” (76 percent) and gender pay gap (22–56 percent) (Figure 7).⁸ In Burkina Faso instead, excessive working hours, living income gap, and occupational health and safety are the main causes of hidden socioeconomic costs. Gender pay gap is the largest social cost item for conventional large producers (57 percent of socioeconomic hidden costs). Child labour is also present in potato production in Burkina Faso, representing 21 and 30 percent of social costs for the sampled “conventional, high-input” and “organic” producers. In Malawi, beyond gender pay gap and living income gap, occupational health and safety is among the largest sources of socioeconomic hidden costs for farmers using local and improved varieties (48 and 50 percent, respectively).

Figure 7. Potato socioeconomic hidden costs: shares of impact categories



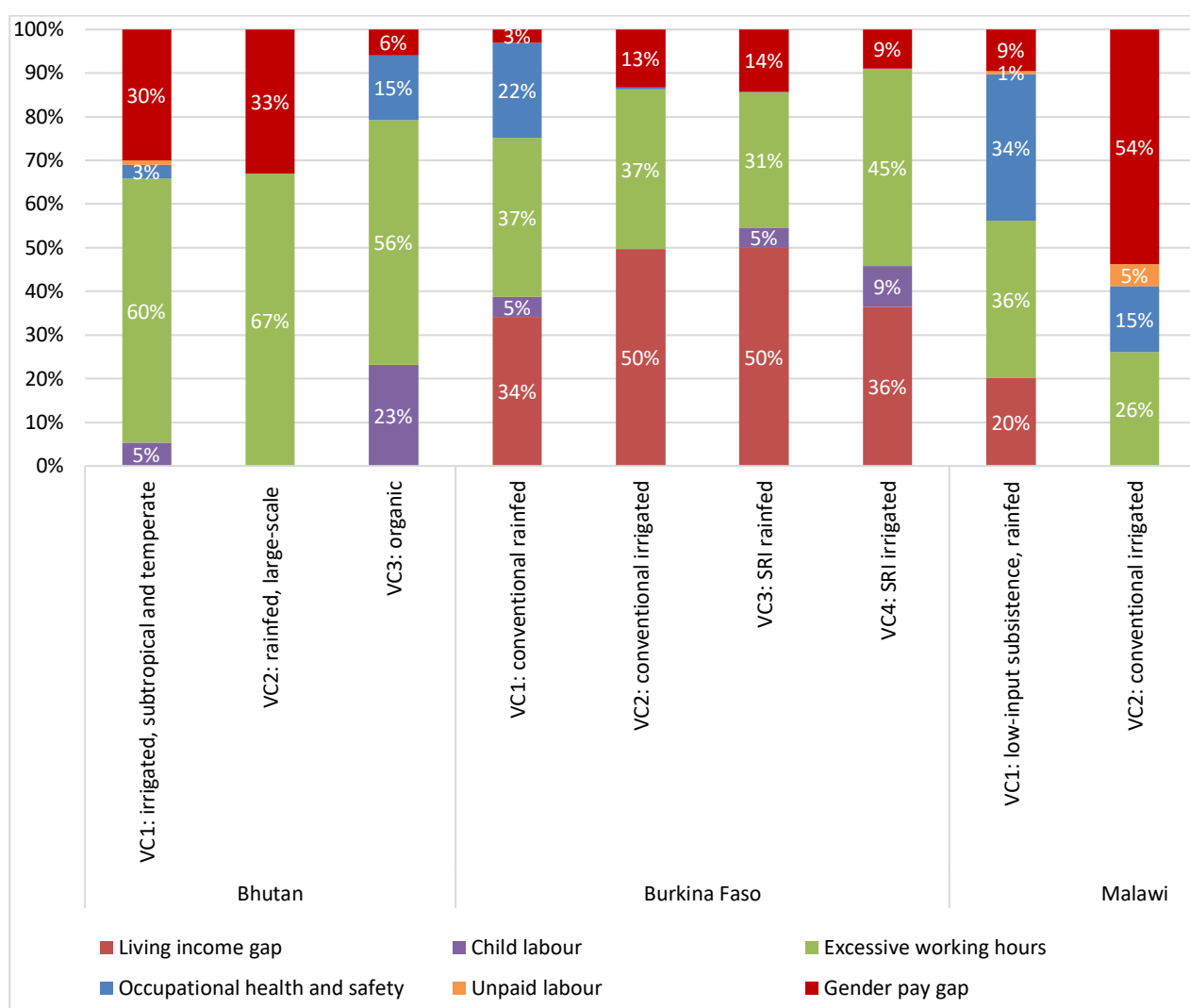
Note: VC = value chain.

Source: Authors' own elaboration.

⁸ For Bhutan an estimate of a living income calculated with the Anker methodology was not available yet. As a proxy, the minimum wage stated by law was used. This estimate should therefore be considered as a lower bound of the living income gap.

The main socioeconomic hidden costs in rice production are caused by excessive working hours and living income gap for Bhutan and Burkina Faso, respectively (Figure 8). Child labour constitutes also a non-negligible share of socioeconomic hidden costs in the case of organic rice production in Bhutan (21 percent), a finding in line with previous estimates at the national level (Khan and Lyon, 2015).⁹ In Malawi, these are caused primarily by excessive working hours, gender pay gap, and occupational health and safety. However, the distribution of costs varies across VCs: conventional irrigated production is characterized by a high share of gender pay gap (54 percent) while low-input subsistence producers by occupational health and safety (34 percent) and excessive working hours (36 percent).

Figure 8. Rice socioeconomic hidden costs: shares of impact categories



Note: VC = value chain.

Source: Authors' own elaboration.

Overall, the results of the socioeconomic assessments highlight the context specificity of hidden costs, both within the same country and across countries. Excessive working hours and living income gap are among the largest hidden costs across almost all the identified value chains. Agricultural work is characterized by long working hours, especially for women (Grassi, Landberg and Huyer, 2015). Poor farmers' need to increase their income often lead them work long hours across multiple jobs or employ family members, including underage children (Garland, 2022).

⁹ According to calculations based on the Bhutan Multiple Indicator Survey (BMIS) in 2010, Khan and Lyon found that overall about 15 percent of children in the age 5–17 years-old were in child labour (Khan and Lyon, 2015, Table 21).

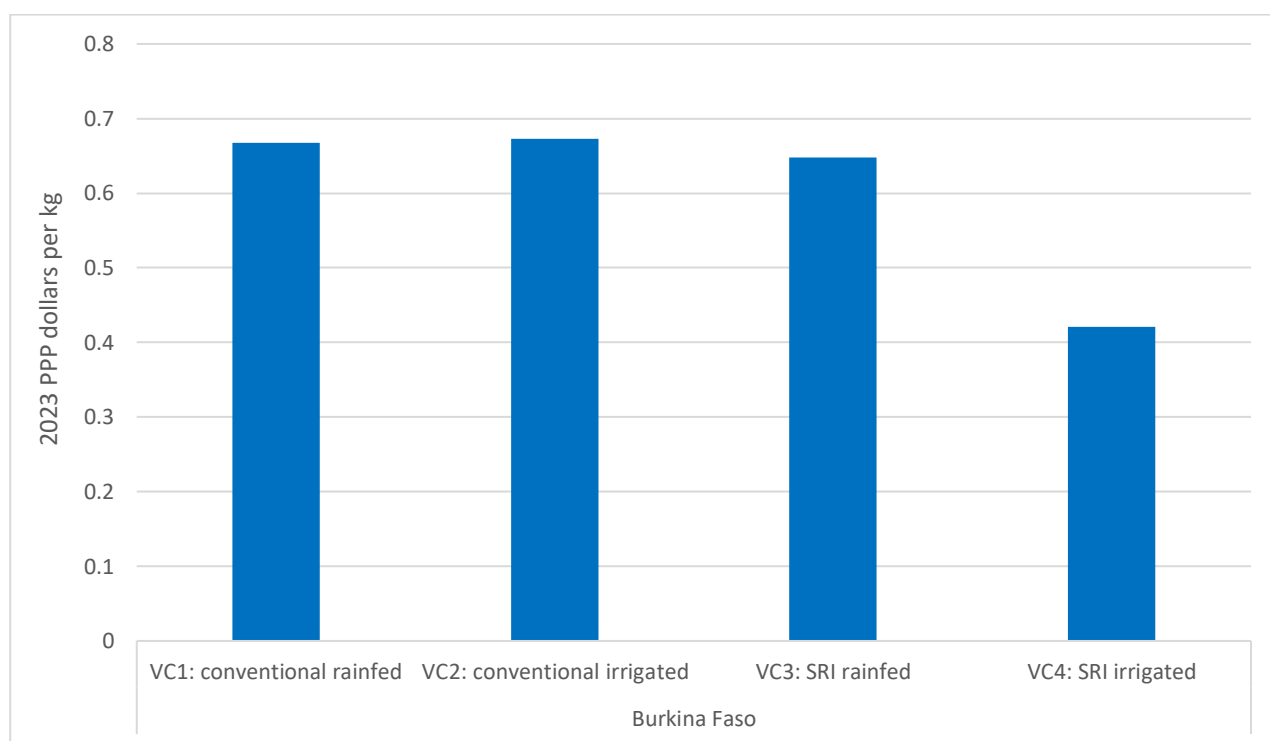
5 Environmental hidden costs in rice production: a comparison of conventional and System of Rice Intensification in Burkina Faso

The study has also attempted to provide insights on how the hidden costs differ between conventional production methods and those that apply improved practices by comparing conventional rice production and the SRI in Burkina Faso. Rice plays a key role for food security, being the main staple grain in West Africa and being the fastest growing staple grain in the rest of sub-Saharan Africa according to Duvallet *et al.* (2021), yet it is also responsible for about 8 percent of global anthropogenic methane emissions (UNEP and CCA, 2021). The SRI is a rice farming methodology introducing low-water, labour-intensive management, and intermitted flooding. It is considered a viable alternative to conventional systems of continuous flooding in order to increase yields while reducing seeds and water consumption as well as reducing emissions from paddies (FAO, 2023c). The study aimed at investigating how those improved practices change the patterns of hidden costs, in particular, whether the environmental costs are effectively reduced.

5.1 Environmental hidden costs

In this part of the study, the environmental hidden costs are calculated both per kg and per hectare for the environmental impacts (Figure 9 and Figure 10). Looking only at environmental costs per kg (mass based functional analysis in LCA) would lead to a bias towards systems that have higher yield and lower hidden costs per kg, overlooking potential absolute increments of certain environmental impacts or ignoring trade-offs (Pérez *et al.*, 2024; Salou, Le Mouél and van der Werf, 2017).¹⁰

Figure 9. Environmental hidden costs per kilogram for Burkina Faso rice production systems

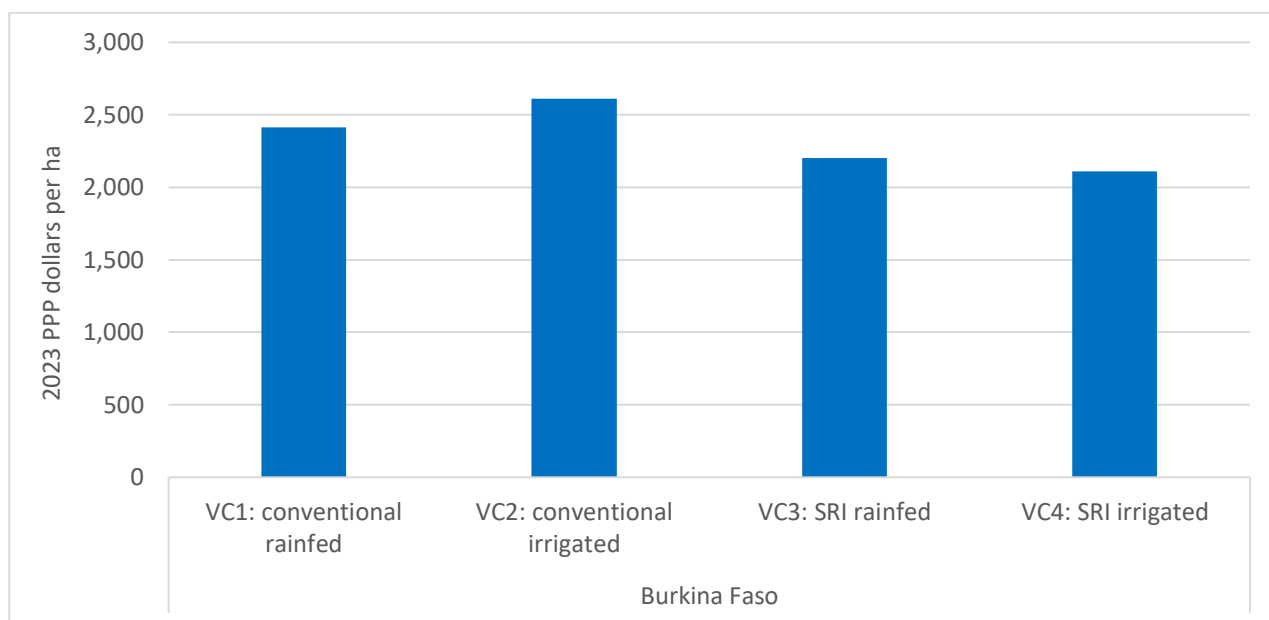


Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

¹⁰ For an example on the cost of ignoring trade-offs, see box 9 on the case of insecticide use in the horn of Africa in *The State of Food and Agriculture 2023* edition.

Figure 10. Environmental hidden costs per hectare cultivated for Burkina Faso rice production systems

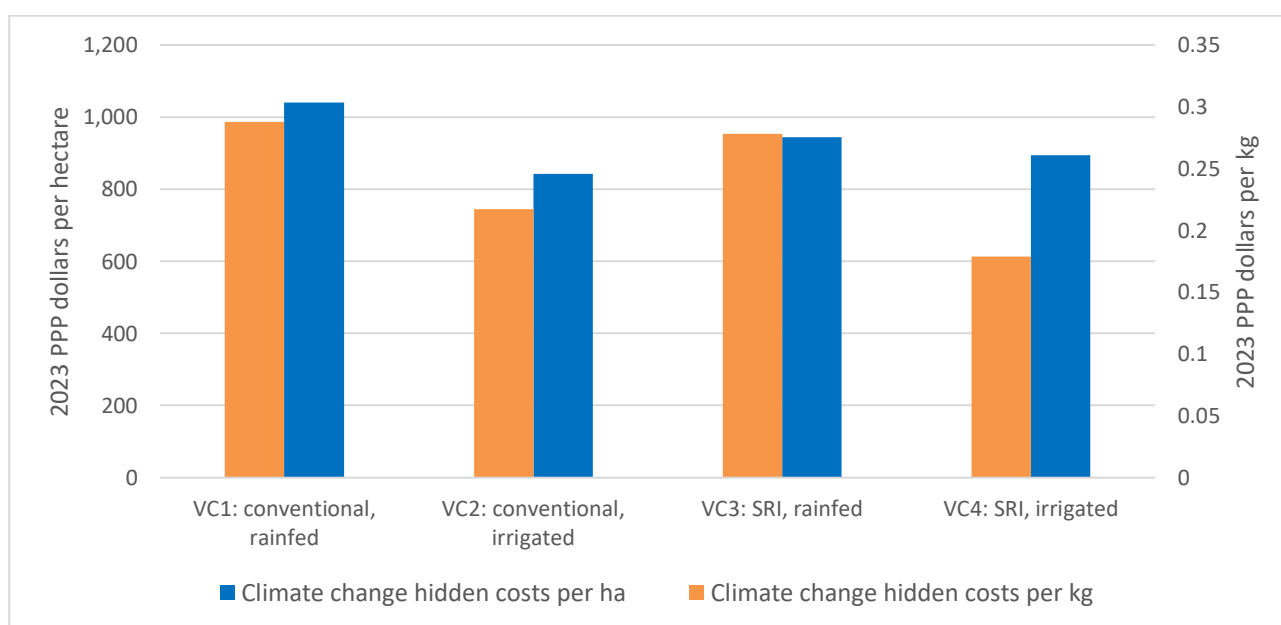


Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

SRI production systems have lower average total environmental hidden both per kg and ha, suggesting that SRI could provide benefits in terms of reduction of potential environmental impacts both in relative and absolute terms. Looking at the single environmental hidden cost indicators – both with kg and ha – confirms lower impacts for SRI for all impact categories, except for climate change. Interestingly, climate change hidden costs per ha are on average slightly higher in SRI irrigated than conventional irrigated: they amount to 894 and 843 2023 PPP dollars per ha, respectively (Figure 11).

Figure 11. Climate change hidden costs of Burkina Faso rice production systems



Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

This result is driven by the fact that a higher share of farmers identified under SRI declared to implement continuous flooding during the rice growing season, despite SRI would require lower use of water through alternate wetting and drying (AWD) periods. Such findings is in line with previous studies documenting different extent of adoption of SRI practices among SRI farmers (Moore *et al.*, 2024; Takahashi, 2013). The choice of keeping a continuous flooding regime among SRI farmers could be affected by the difficulty of creating drainage channels to implement AWD and the difficulty to drain water from paddies during the rainy season (Johnson, Vijayaragavan and Singh, 2012; Moore *et al.*, 2024).

5.2 Discussion

These results provide important insights in the context of informing decision-making processes. First, such targeted TCA assessments can help stakeholders in identifying major sources of hidden costs and understanding how these might change across different commodities or across different management practices for the same commodity. Secondly, the results of the rice systems comparison underline the importance of looking at indicators from both an eco-efficiency (results expressed per unit of product) and land use perspectives (per unit area). Focusing only on a single indicator – for instance GHG emissions per kg of product (emission intensity) and not per ha (total emissions) – could hide important information, potentially leading to prioritization of a specific management practice, which might not be suitable to reach the objective that decision-makers seek to achieve (e.g. reduction of total GHG emissions). As a result, countries that are planning to implement a specific practice for a specific commodity to advance their sustainability agenda could gain useful insights by looking at these different metrics and better tailor their policy response. In case of promoting SRI among climate mitigation and adaptation practices (SRI-2030, 2023), they could benefit from additional useful information by conducting TCA assessments of the main rice cultivation practices using both types of indicators to carefully delineate the flooding regime.¹¹

5.3 Insights from the data collection process

TCA studies generally require a considerable amount of data across many dimensions, which can be obtained through i) primary data collection (e.g. surveys and physical measurement), ii) secondary data and iii) estimated data using models (FAO, 2023a). The majority of data used in this study come from surveys. This approach allows for gathering suitable information for the assessment, while at the same time provides an opportunity to familiarize local stakeholders, researchers, and farmers with TCA purposes and concepts. However, the following elements should be considered when conducting a TCA study using surveys:

- A survey-based TCA analysis is subject to the well-known agricultural data collection errors such as measurement and non-response errors resulting from ill-timing of the survey instrument, farmers not keeping track of expenses or inputs, use of terms which are not understood, insufficient interviewer training, measurement errors related to productivity and cultivation area from self-reported data (Ayalew *et al.*, 2024; Carletto, Dillon and Zezza, 2021; FAO, 2016).
- As TCA data needs cover many dimensions, with questionnaires requiring detailed information on inputs use quantities, prices and labour conditions, the time to conduct an interview could be substantial depending on the desired level of detail. In addition, TCA requires information on activities conducted over different phases of production, from sowing to harvesting. As a result, respondents' answers might not be accurate and estimates based on these data subject to recall bias, which can increase with the length of recall especially for variables such as land, labour, fertilizer, and production (Wollburg, Tiberti and Zezza, 2021). To reduce recall bias some approaches could be explored for future survey-based TCA, including two field visits (one in the post-planting and one in the post-harvest period) as well as the use of diaries (Wollburg, Tiberti and Zezza, 2021).

¹¹ The choice on whether to consider emission intensity or total emissions eventually depends on the policy objective pursued (FAO, 2023c). In general, actors planning to conduct TCA studies should be aware on the role of different functional units and their limitations choosing the most appropriate (de Adelhart Toorop *et al.*, 2023). Additional information on the use of LCA for methane mitigation strategies can be found in chapter 8 of the report "Methane emissions in livestock and rice systems – Sources, quantification, mitigation and metrics (FAO, 2023c).

- Quantities could be difficult to estimate for farmers, especially for those items that are not sold nor bought, including water for irrigation or the amount of residues after harvest.
- Farmers might report the quantities of inputs using local units of measure (e.g. number of ox-cart, bags, buckets). To facilitate data collection processes, questionnaires should be adapted to include answer options with local units, or the enumerators should be properly trained on conversion factors from local units to the units used in the questionnaire (e.g. kg corresponding to a bucket of rice husk).
- Given the wide scope of the questionnaires, adequate training for enumerators is critical to reduce the number of errors in filling the survey and to optimize the ensuing data elaboration.

6 Limitations

The findings of this study are subject to some important limitations. The results are based on a small sample size, building on convenience sampling given the purpose of the analysis, and should be therefore treated with caution limiting the conclusions to illustrative purposes. Second, given that the analysis was mainly based on primary data collected with comprehensive questionnaires, some variables, especially those difficult to be estimate by farmers, are likely affected by measurement errors. Finally, adopting an LCA approach to measure environmental impacts implies that the results are also affected by limitations intrinsic to LCA modelling such as uncertainties in primary data and in the way impact pathways are modelled or the incomplete coverage of different impacts of agrifood systems (Deconinck and Toyama, 2022).

Future analyses could explore mixed-methods approaches, for example using primary data collection with a representative sample and applying direct measurement in selected farms to proxy for variables otherwise difficult to estimate (e.g. water use, electricity consumption during processing). Such approaches have the potential to reduce the time required to complete an interview and, at the same time, to improve the accuracy of collected data.

7 Conclusions

The following key findings emerge from our pilot TCA study.

Overall, the socioeconomic hidden costs represent the largest share of hidden costs. These are mainly due to low incomes and excessive working hours, for both potato and rice systems in all the three countries analysed. Additionally, gender pay gap and injuries represent an important source of hidden costs in Malawi.

The share of each socioeconomic hidden cost category vary widely both across different value chains in the same country and across different countries. These findings highlight the importance of considering the context specificity of TCA results in decision making processes and call for caution in conducting cross-country comparison of hidden costs of the same crop. Results from a TCA assessment of a crop in one specific country should therefore not be taken as a proxy of main externalities of the same crop in a different country.

The key environmental hidden costs in potato production are human toxicity, particulate matter, and climate change. For rice, they include climate change mainly due to methane emissions from paddies, followed by human toxicity. To minimize the environmental hidden costs associated with rice production, it is crucial that TCA assessments consider the varying water management regimes of rice paddies.

The findings from comparing conventional and SRI methods in Burkina Faso highlight the need to analyse environmental hidden costs using both per kilogram and per hectare metrics. This dual approach provides a more comprehensive understanding that can guide policy decisions, focusing not just on total hidden costs but also on individual cost indicators across both metrics. In fact, using only one of the two metrics could favour a management practice that would not be optimal from perspective of reaching established policy objectives.

Finally, the data collection process highlighted the wide range of data required to conduct TCA and the related length of questionnaire as well as the importance to adapt the questionnaire to local context (no one-size-fits-all TCA questionnaire). To minimize measurement error, future analyses could consider mix-methods approaches, in which standard surveys are combined with other approaches such as using targeted directed measurement for variables difficult to quantify by farmers (e.g. water use, biomass residues), two-time field visits or the complementary use of both standard TCA survey and household diaries.

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Annex 1. Hidden costs

Table A1. Potato value chains: environmental and socioeconomic hidden costs

Country	Value chain	Environmental	Socioeconomic	Total
	2023 PPP dollars per kg			
Bhutan	VC1: smallholder	0.44	0.57	1.01
	VC2: commercial	0.21	1.20	1.41
	VC3: organic	0.53	1.95	2.48
Burkina Faso	VC1: conventional small and medium	0.56	0.89	1.45
	VC2: conventional large	0.53	0.19	0.72
	VC3: conventional high-input	0.55	1.25	1.80
	VC4: organic	0.47	35.76	36.23
Malawi	VC1: low-input subsistence farmers, rainfed	0.40	2.22	2.62
	VC2: local varieties, minimum fertilizer, extensive irrigation	0.28	1.27	1.56
	VC3: improved varieties, minimum fertilizer, irrigated	0.21	1.05	1.26

Note: VC = value chain.

Source: Authors' own elaboration.

Table A2. Rice value chains: environmental and socioeconomic hidden costs

Country	Value chain	Environmental	Socioeconomic	Total
	2023 PPP dollars per kg			
Bhutan	VC1: irrigated, subtropical and temperate	0.86	3.09	3.95
	VC2: rainfed, large scale	0.68	7.47	8.14
	VC3: organic	0.86	3.28	4.15
Burkina Faso	VC1: conventional, rainfed	0.67	2.15	2.81
	VC2: conventional, irrigated	0.67	0.81	1.49
	VC3: SRI, rainfed	0.65	0.73	1.38
	VC4: SRI, irrigated	0.42	1.01	1.44
Malawi	VC1: low-input subsistence, rainfed	0.96	5.02	5.97
	VC2: conventional, irrigated	0.67	2.13	2.80

Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

Table A3. Rice value chains in Burkina Faso: environmental hidden costs per kg

	VC1: conventional, rainfed	VC2: conventional, irrigated	VC3: SRI rainfed	VC4: SRI irrigated
Acidification	7.35E-03	9.35E-03	7.37E-03	4.80E-03
Climate change (GWP100)	2.88E-01	2.17E-01	2.78E-01	1.79E-01
Ecotoxicity, freshwater	1.24E-03	3.15E-03	7.19E-04	1.66E-03
Eutrophication, freshwater	3.65E-04	3.73E-04	1.98E-04	1.94E-04
Eutrophication, marine	5.56E-02	6.99E-02	5.74E-02	3.55E-02
Human toxicity, carcinogenic	9.05E-03	1.18E-02	8.53E-03	7.16E-03
Human toxicity, non-carcinogenic	1.85E-01	2.08E-01	1.83E-01	1.10E-01
Ionising radiation (human health)	1.32E-05	1.59E-05	6.59E-06	8.46E-06
Land use	1.42E-02	1.49E-02	1.66E-02	1.06E-02
Ozone depletion	7.07E-07	2.11E-06	6.30E-07	1.45E-06
Particulate matter	8.88E-02	1.16E-01	8.30E-02	6.24E-02
Photochemical ozone formation	2.68E-03	3.26E-03	1.78E-03	2.38E-03
Resource use, fossils	6.06E-03	6.86E-03	2.94E-03	3.57E-03
Resource use, minerals and metals	1.06E-05	1.15E-05	4.78E-06	5.92E-06
Water use (user deprivation potential)	9.57E-03	1.16E-02	7.96E-03	4.19E-03
Total	6.68E-01	6.73E-01	6.48E-01	4.21E-01

Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

Table A4. Rice value chains in Burkina Faso: environmental hidden costs per hectare

	VC1: conventional, rainfed	VC2: conventional, irrigated	VC3: SRI rainfed	VC4: SRI irrigated
Acidification	26.56	36.28	25.05	24.01
Climate change (GWP100)	1 040.00	842.58	945.09	894.15
Ecotoxicity, freshwater	4.47	12.23	2.44	8.29
Eutrophication, freshwater	1.32	1.45	0.67	0.97
Eutrophication, marine	200.80	271.43	194.98	177.77
Human toxicity, carcinogenic	32.71	45.82	28.97	35.84
Human toxicity, non-carcinogenic	667.55	807.06	622.76	550.79
Ionising radiation (human health)	0.05	0.06	0.02	0.04
Land use	51.34	57.95	56.42	53.08
Ozone depletion	0.00	0.01	0.00	0.01
Particulate matter	320.82	452.14	281.76	312.29
Photochemical ozone formation	9.69	12.65	6.03	11.89
Resource use, fossils	21.91	26.63	9.98	17.89
Resource use, minerals and metals	0.04	0.04	0.02	0.03
Water use (user deprivation potential)	34.57	45.13	27.02	20.97
Total	2 411.83	2 611.47	2 201.22	2 108.02

Notes: SRI = System of Rice Intensification; VC = value chain.

Source: Authors' own elaboration.

Annex 2. Environmental impact assessment methodology

Country and value chain specific data used for modelling the foreground system is based on input and output data of national potato and rice value chains collected by the respective country teams. Nitrous oxide (N₂O) and phosphorus emissions due to the use of fertilizers are modelled according to the recommendations of Pant and Zampori (2019). N₂O emissions from harvest residues are modelled according to IPCC (IPCC, 2019). Heavy metal emissions to agricultural soils caused by the application of fertilizers and pesticides are estimated based on Freiermuth (Freiermuth, 2006). It is assumed that 90 percent of the pesticides are released to agricultural soils, 9 percent to air and 1 percent to water (Zampori and Pant, 2019). Background data on emissions associated with production of the input substrates such as fertilizers or pesticides are based on the *ecoinvent* database 3.9 using the cut-off system model (Wernet *et al.*, 2016). In the current study market datasets are used in order to include average transport impacts (Wernet *et al.*, 2016). The software openLCA 1.11 is applied for the modelling and the calculation of the impacts.

For the impact categories fossil depletion, metal depletion as well as water depletion no monetization factors are available. The monetization factors were only available for the year 2017. To account for inflation, they were adapted for the year 2023 based on the approach described in Arendt *et al.* (2020).

As biodiversity impacts differs largely from farm to farm and on-farm biodiversity assessments are very time-consuming, it is beyond the scope of this study. Impacts of the value chains on natural biodiversity are, however, assessed based on the environmental indicators freshwater ecotoxicity, freshwater eutrophication, marine ecotoxicity, marine eutrophication, terrestrial acidification, terrestrial ecotoxicity and water depletion.

Other ecosystem services such as soil erosion prevention, water filtration, moderation of extreme events (e.g. floods) and air quality regulation are very site specific and require field-specific data. Obtaining these data is, however, as well very time and resource intensive, and most important only valid for the specific location of an individual farm. Neither the expression of the value as an average nor as min-max value would be meaningful for a true cost assessment. Thus, these ecosystem services are excluded from this study.

Annex 3. Socioeconomic impact assessment methodology

Living income gap

The living income gap was calculated adapting the approach suggested by the *True cost accounting agrifood handbook* (True Cost Initiative, 2022), using the following formula for each management practice:

$$LIG_{c,i} = \frac{\frac{I_c}{I_{tot}} \times (LI_i - (I_{tot} - Cost_c))}{OP_c}$$

Where:

$LIG_{c,i}$: is the living income gap for crop c and farming practice i , representing the additional income obtained from crop c needed to reach a living wage;

LW : is the reference Living Income for the country in scope obtained for Burkina Faso and Malawi from the Global Living Wage Coalition¹² and for Bhutan from the WageIndicator Network¹³.

I_c : is the income obtained from crop c ;

I_{tot} : is the median total income of the household;

I_c : is the income obtained from crop c ;

$Cost_c$: are the expenses related to production of crop c (excluding labour costs);

OP_c : average kg of crop c per farm and year.

Child labour

The impact indicator child labour was calculated based on the *True cost accounting agrifood handbook* (True Cost Initiative, 2022) and adapting their formula as follows:

$$Child\ labour_{c,i} = MF_{CL} \times \left(\left(\frac{child_work_hours_{c,i} - 560}{2\,240} \right) \times 0.5DALY \times \frac{1}{OP_{c,i}} \times \frac{I_{c,i}}{I_{tot}} \times \frac{HH_{chil>560h}}{HH_{tot}} \right)$$

Where:

$Child_work_hours$: average working hours per child for crop c and farming practice i ;

MF_{CL} : monetization factor for child labor¹⁴ (2023 PPP dollars/DALY);

I_c : is the income obtained from crop c ;

I_{tot} : is the total income of the household;

HH_{chil} : households with children working more than 560 hours per year;

HH_{tot} : total number of households;

OP_c : average kg of crop c per farm and year.

¹² "The net annual income required for a household in a particular place to afford a decent standard of living for all members of that household. Elements of a decent standard of living include: food, water, housing, education, healthcare, transportation, clothing, and other essential needs including provisions for unexpected events." (Global Living Wage Coalition, 2024).

¹³ <https://wageindicator.org/salary/minimum-wage/bhutan>

¹⁴ 145 531 2023 PPP dollars obtained from the initial value of EUR 80 000 (2017) and adjusting for inflation using the CPI for the European Union 28 (prc_hicp_midx) from Eurostat and PPP conversion factor for Euro area (19 countries) from the OECD Annual Purchasing Power Parities and exchange rates (OECD, 2024).

Excessive working hours

The impact indicator excessive working hours was calculated based on the *True cost accounting agrifood handbook* (True Cost Initiative, 2022) and adapting their formula as follows:

$$Excessive\ working\ hours = MF_{EW} \times \left(\left(\frac{Avg_{workhours} - 2\ 240h}{2\ 240h} \right) \times 0.5DALY \times \frac{1}{OP_c} \times \frac{I_c}{I_{tot}} \times HH_{>2\ 240h} \right)$$

Where:

Avg_workhours: is the average annual working hours;

0.5DALY: disability adjusted life years in 2023 PPP dollars;

MF_{CL}: monetization factor for excessive working hours (2023 PPP dollars/DALY);

OP_c: average kg of crop *c* per farm and year;

I_c: is the income obtained from crop *c*;

I_{tot}: is the total income of the household;

HH_{>2 240h}: share of household member [in %] working more than 2 240 hours per year.

Occupational health and safety

The impact indicator occupational health and safety accounts for the health impact from work-related injuries, illness and death of workers (True Cost Initiative, 2022) and was calculated using the following formula:

$$\begin{aligned} Occupational\ health\ and\ safety \\ = MF_{OHS} \\ \times \left\{ \left[\sum_{j=0}^q (Illness_j \times DW_j) \times (365) + \sum_{k=0}^n (Injury_k \times DW_k) \times (LE - A_k) \right. \right. \\ \left. \left. + \sum_{l=0}^m (Fatality_l \times DW_l) \times (LE - A_l) \right] \times DALY \times \frac{1}{Workers_{tot}} \times \frac{1}{OP_c} \right\} \end{aligned}$$

Where:

MF_{OHS}: monetization factor for occupational health and safety (2023 PPP dollars/DALY);

Illness_j: number of days absent from work due to illness per worker *j* per year;

Injury_j: number of injuries per injury type *k* per year;

Fatality_j: number of fatalities per worker *l* per year ;

DW_j: disability weight (DW) assigned to death is equal to 1. DWs were taken from the *Global Burden of Disease* (Vos *et al.*, 2020);

DW_k: disability weight (DW) of the injury type *k*;

DW_l: disability weight (DW) of the injury type *l*;

Q: number of days absent from work due to illnesses per year;

N: number of injuries per year;

M: number of fatalities per year;

LE: national life expectancy (years);

A: age of worker (years);

Workers_{tot}: number of household members and non-household agricultural workers;

OP_c: average kg of crop *c* per farm and year.

Unpaid labour

The indicator forced labour proxies for regular work without remuneration of non-household members and is calculated using the following equation:

$$Unpaid\ labour = Avg_wkh_nonpaid \times PL \times \frac{1}{HH_{tot}} \times \frac{1}{OP_c}$$

Where:

Avg_wkh_nonpaid: average annual working hours without remuneration;

PL: international poverty line from World Bank;

HH_{tot}: total number of farms;

OP_c: average kg of crop *c* per farm and year.

Gender pay-gap

The indicator gender pay-gap is calculated using the following formula adapted from True Cost Initiative (2022):

$$Gender\ pay\ gap = (Avg_{income_h} - Avg_{income_l}) \times \frac{1}{OP_c} \times Work_hours_l$$

Where:

Avg_income_h: annual average income of the sex with the higher remuneration (currency/hour);

Avg_income_l: annual average income of the sex with the lower remuneration (currency/hour);

OP_c: average kg of crop *c* per farm and year;

Work_hours_l: average working hours of sex with the lower remuneration.

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